

Magnetic Resonance Imaging (MRI) Safety

Policy Number	POL-SAF-071	Owner / Department	Radiology / Safety
Effective Date	2023-11-01	Version	2.3
Last Revised	2026-04-08	Next Review	2028-04-08
Approved By	Radiation Safety Committee	Applies To	All staff, providers, patients, visitors, contractors, and emergency responders entering or working in the MRI suite

1. Purpose and Scope

This policy establishes a four-zone access-control and screening program for the MRI environment to prevent ferromagnetic-projectile, burn, implant, acoustic, and cryogen-quench hazards. It follows the American College of Radiology (ACR) Manual on MR Safety and applies to the entire MRI suite at all times — the static magnetic field is **always on**, even when no scan is in progress.

The program is led by the **MR Medical Director** and the **MR Safety Officer (MRSO)**, supported by MR Safety Experts, with authority to restrict access.

2. ACR Safety Zones

Access is controlled across four zones of increasing field strength and restriction. No one enters Zone III or IV without completing MR screening.

Zone	Area	Access
Zone I	General public areas outside the MR department	Open; no screening required
Zone II	Reception, changing, and initial screening interface between public and controlled space	Patients under staff supervision
Zone III	Controlled area around the console/magnet, where unscreened ferromagnetic items are dangerous	Screened, MR-trained personnel only
Zone IV	The magnet (scanner) room itself	Screened persons under direct MR-staff supervision

IMPORTANT — No ferromagnetic item may cross into Zone IV. Oxygen cylinders, IV poles, wheelchairs, tools, and pens must be MR-conditional or MR-safe and screened. A ferromagnetic object becomes a lethal projectile near the bore.

3. Device and Implant Labeling

Every device, implant, and item brought near the magnet must carry a known MR classification before entry:

- **MR Safe** — poses no known hazard in any MR environment (non-conducting, non-metallic, non-magnetic).

- **MR Conditional** — safe only within the specified conditions (field strength, gradient, and RF limits) documented by the manufacturer.
- **MR Unsafe** — poses an unacceptable hazard and is prohibited from Zone IV.
- Implant eligibility is verified against manufacturer MR conditions and documented by the MRSO/radiologist before scanning.

4. Screening and Emergency Response

1. Complete a written MR screening form for every patient and any person entering Zone III/IV; reconcile metal implants, devices, and prior injuries.
2. Perform a verbal re-check and ferromagnetic detection at the Zone III/IV threshold.
3. Remove external ferromagnetic items and use only MR-conditional ancillary equipment inside Zone IV.
4. For cardiac arrest, remove the patient from Zone IV before resuscitation; code teams do not bring ferromagnetic equipment into the magnet room.
5. For a cryogen quench (oxygen-displacement and frostbite risk), evacuate the room, do not re-enter until cleared, and press the magnet-rundown/quench button only when a projectile traps a patient against the bore.

5. Severity & Reporting

Classify and notify per **POL-RM-013**. A ferromagnetic projectile event causing serious injury, or an unintended quench with patient/staff harm, is a **SEV-1**; a projectile event or burn without serious injury is a **SEV-2**; a screening breach with no harm is a **SEV-3 / Near Miss**.

Severity	Notify	Deadline
SEV-1 (catastrophic / sentinel)	MR Medical Director, MRSO, Risk Management, administrator on call	Within 1 hour
SEV-2 (major)	MRSO and Radiology manager	Within 4 hours
SEV-3 / Near Miss (minor)	MRSO via electronic event report	Within 24 hours

If a projectile involved a medical device, also follow the device-reporting policy **POL-SAF-073**; staff injuries are cross-referenced to **POL-EH-001**.

6. References

- ACR Manual on MR Safety — zones I–IV and MR Safe / MR Conditional / MR Unsafe labeling.
- POL-RM-013 — Incident Reporting and Event Severity Classification.
- POL-SAF-073 — Medical Device Malfunction, Recall, and FDA Reporting.
- POL-EH-001 — Bloodborne Pathogen Exposure and Needlestick.